COMMERCIAL AVIATION · INDIA AND THE GULF

India's Wide-Body Window

Where should Indian carriers deploy their next 100 long-haul aircraft, and can the India-Gulf corridor absorb them?

THE ANSWER

Compete with the Gulf hubs. Do not fly more aircraft to them. The corridor carries half of India's international traffic and is four times the entire direct Europe market, but a fifth of it is passengers going somewhere else entirely, and what treaty room remains in the Gulf would absorb only about four per cent of the aircraft on order. The wide-bodies win that traffic by flying past the Gulf, not to it. Europe first, North America second.

CLIENT

IndiGo, network and fleet strategy

THE DECISION

Where 60 A350-900s on firm order go first, and what to do with 40 unconverted purchase rights

HORIZON

Deployment through 2030

AGAINST

Air India, 80 wide-bodies on firm order

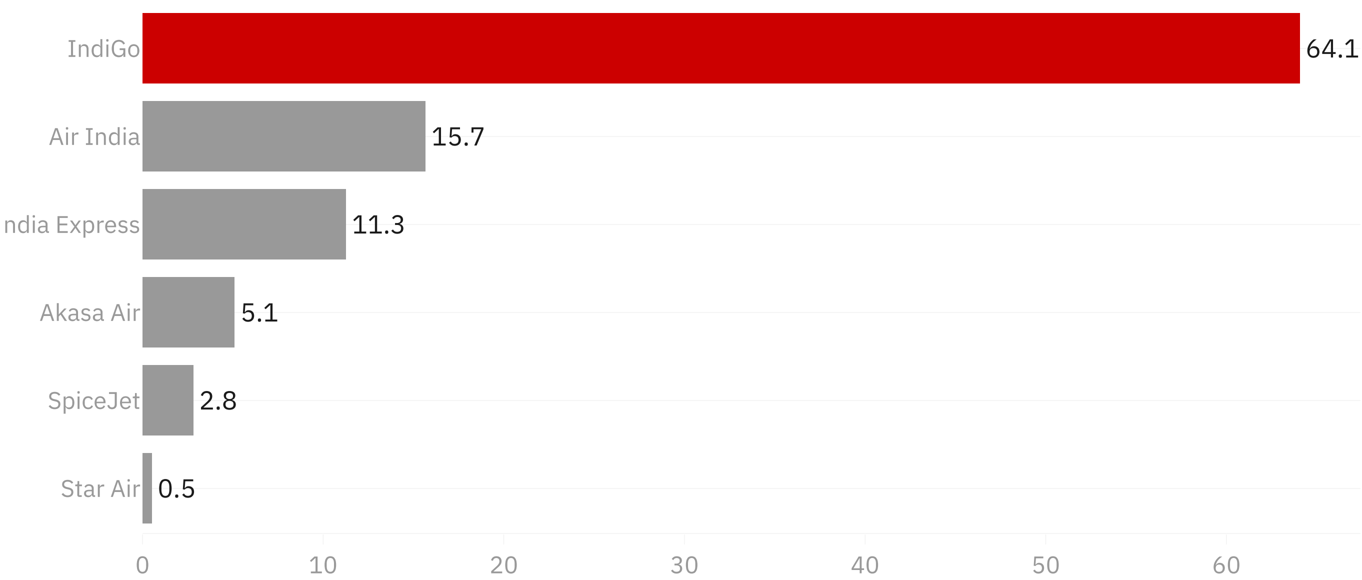
The base is already won

IndiGo carries roughly two thirds of every domestic passenger in India. That position generates the cash and the feed network any long-haul expansion has to be built on.

Domestic is not the problem. It is the platform.

IndiGo owns two thirds of the domestic market that funds the wide-bodies

Share of scheduled domestic passengers, per cent, 2025



Source: DGCA traffic statistics, via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

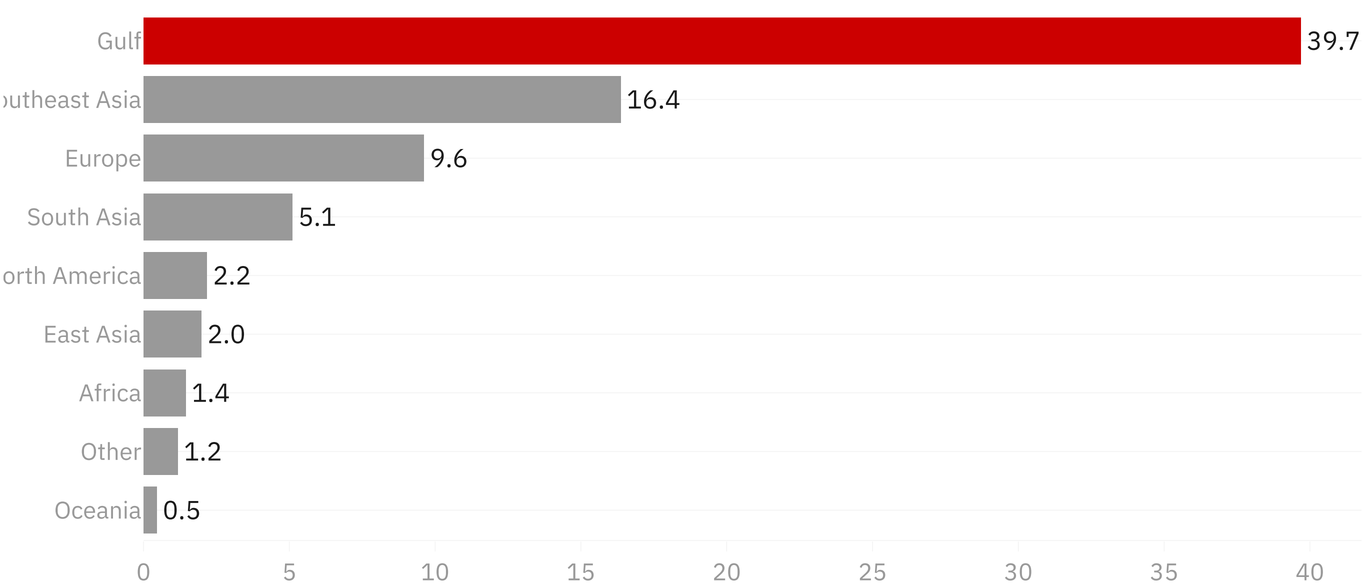
The prize is the Gulf, by a distance

Half of India's international traffic touches a Gulf point. The corridor is four times the size of India's entire direct market to Europe, and larger than Europe, North America, East Asia, Africa and Oceania combined.

Any conversation that starts with London and New York has already missed where the volume is.

The Gulf corridor is four times the size of India's entire direct Europe market

India international sector passengers by destination region, millions, 2025



Source: DGCA traffic statistics, via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

And the passenger vanishes there

India's own statistics record the first foreign point and nothing beyond it. A passenger flying Delhi to Dubai to London is counted as a passenger to the United Arab Emirates.

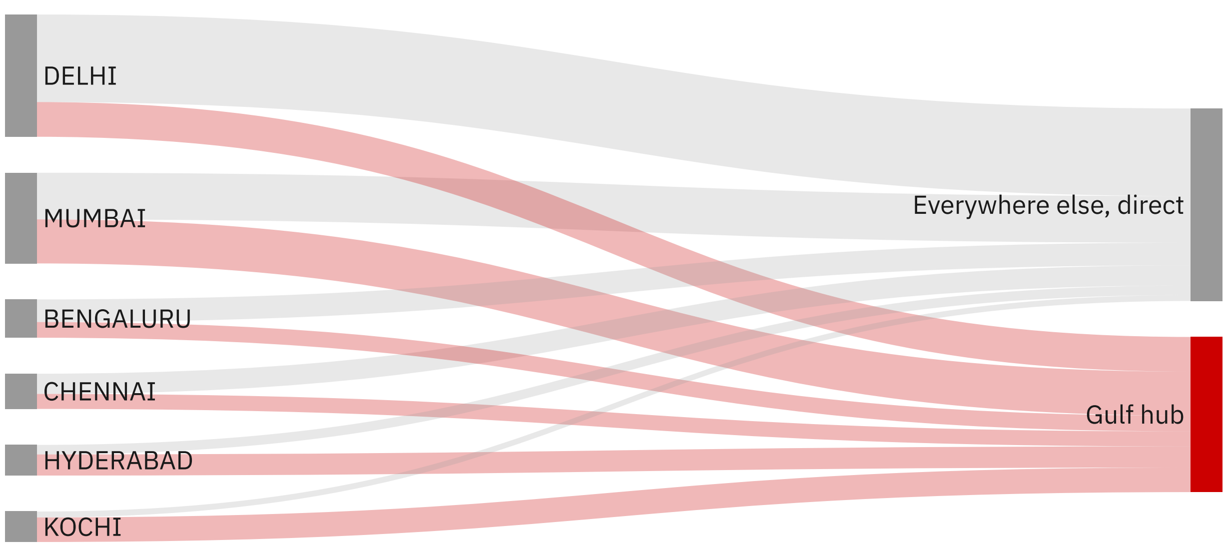
That blind spot is not a data problem to apologise for. It is the finding. What the Gulf hubs do with those passengers after they land is where the margin goes, and India cannot see it.

Kochi sends more passengers to Gulf hubs than to the entire rest of the world combined.

> **PIVOT 3**
>
> This chart was wrong for weeks. DGCA writes `ABUDHABI` where the code looked for `ABU DHABI` , so 5.0M passengers a year, a fifth of the entire Gulf hub flow, sat in the "everywhere else" bucket. All 72 tests passed the whole time, because a wrong bucket is still a valid bucket. How this analysis changed

India's own statistics lose sight of the passenger at the Gulf hub

Passengers from India's largest international gateways, 2025. DGCA records the first foreign point only, so what ha



Source: DGCA traffic statistics, via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

What the blind spot is worth

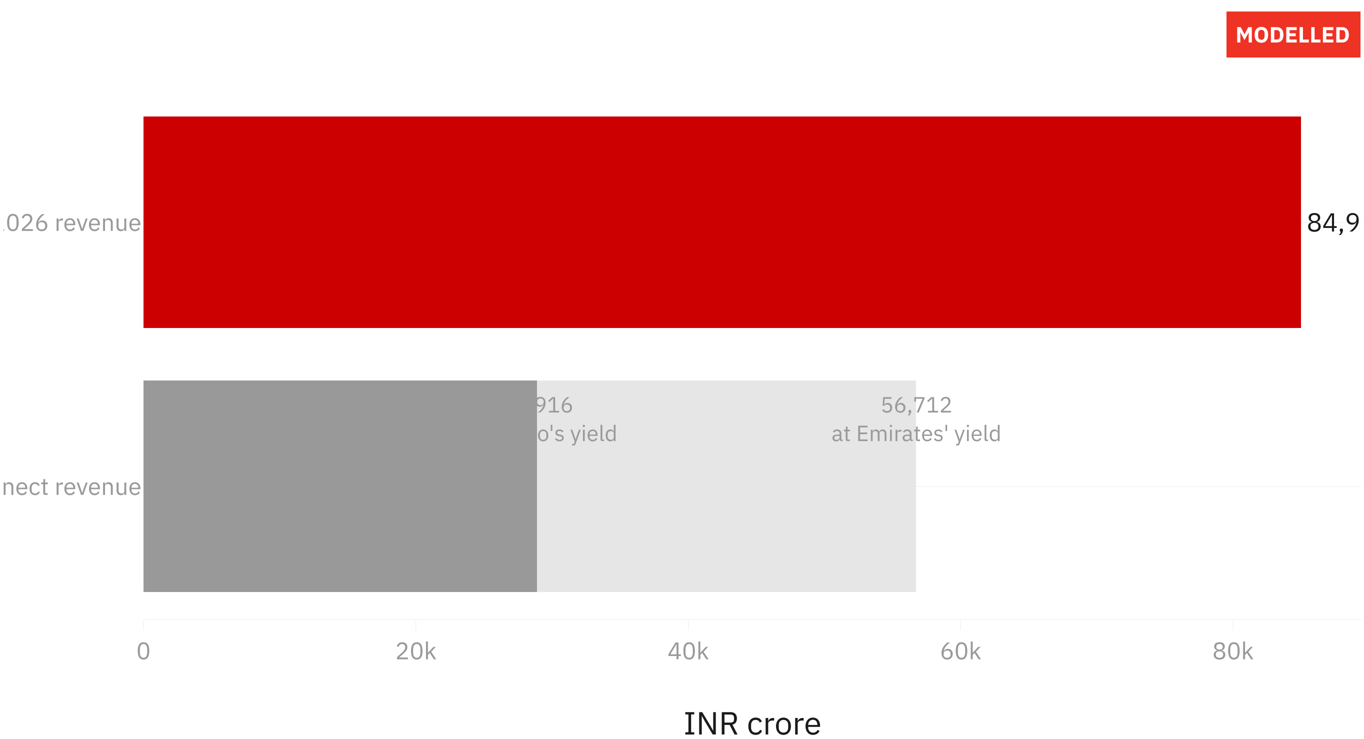
India's own statistics put the Gulf at 51 per cent of international sectors. Origin-destination figures, which follow the passenger to where they were actually going, put it near 40. The eleven point difference is roughly eight and a half million people a year whose journey begins in India, ends in Europe or North America, and is sold by a Gulf carrier.

Priced across that journey at the only two yields this project has verified, its own and Emirates', the contested revenue runs between 28,900 and 56,700 crore rupees. A third to two thirds of IndiGo's entire annual revenue, in a pool it does not currently compete for.

Read it as the size of the contested pool, not a prize anyone captures whole. And note the weakest link is named: the 40 per cent cannot be verified, because IATA sells that data and publishes no free table. It is gated as unverifiable and everything drawn from it is reported as a band.

The passengers India cannot see carry INR 28,916 to 56,712 crore, a third to tv

8.5M passengers a year, the gap between the Gulf's 50.9% of India's international sectors and roughly 40% of true c



Source: Unit costs and yields from IndiGo FY2026 and Emirates 2025-26 primary filings, verified in data/manual/assumptions.csv

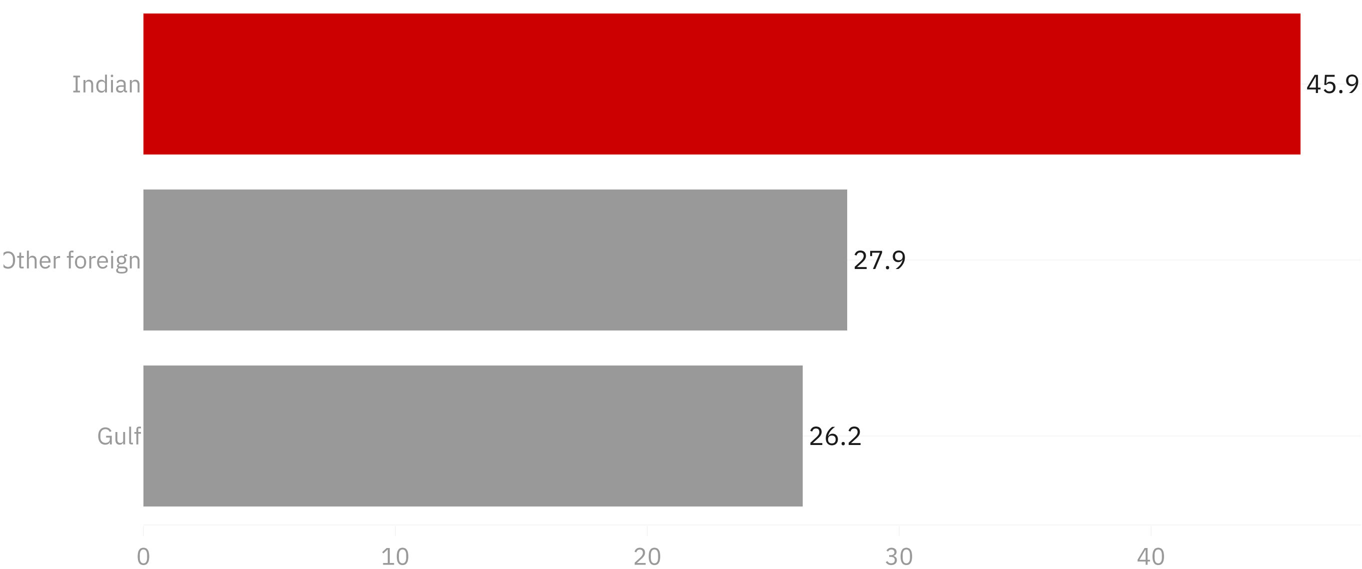
Foreign carriers still fly the majority

Indian carriers account for fewer than half of India's international passengers. Gulf carriers take a quarter.

This is a market where the home team is still the minority shareholder.

Indian carriers fly fewer than half of India's own international passengers

Share of India international sector passengers by carrier home region, per cent, 2025



Source: DGCA traffic statistics, via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

But the trend belongs to the Indian carriers

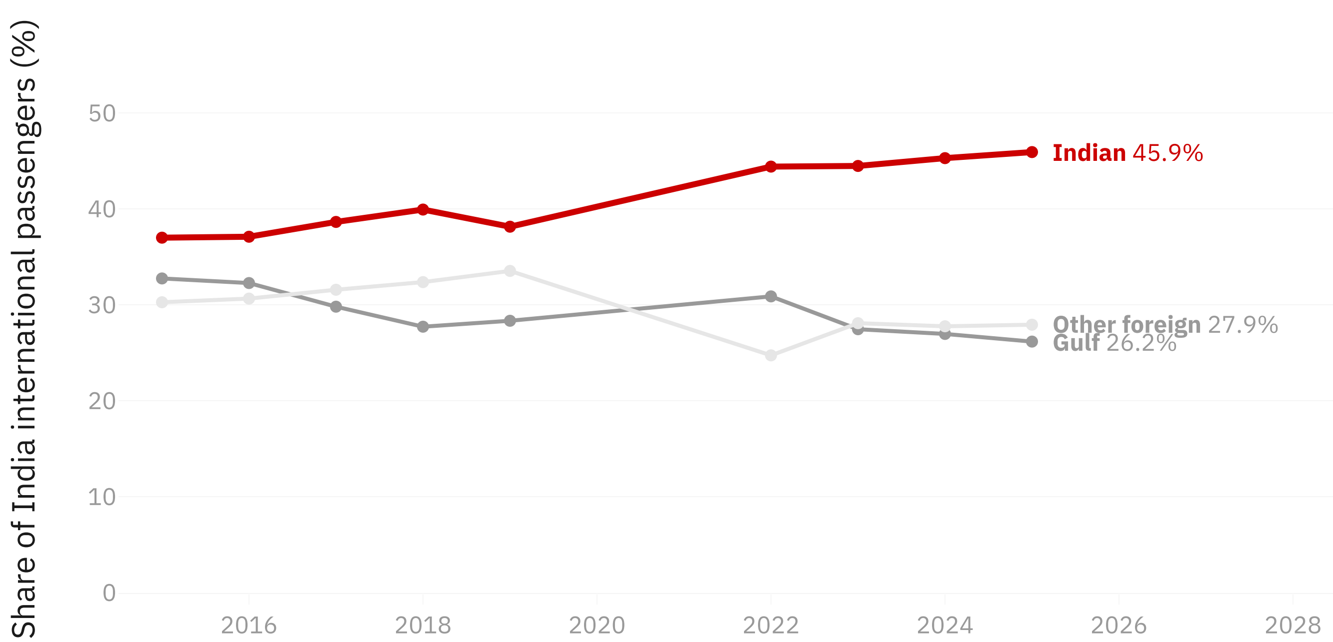
The obvious version of this case is that India is losing its own market and needs rescuing. The data says the opposite. Indian carriers have gone from 37 per cent of international traffic in 2015 to 46 per cent in 2025, while the Gulf carriers have fallen from 33 per cent to 26.

That matters, because a recommendation built on a false premise is worth nothing. The gap is closing on its own. What has not closed is where it is closing: the share taken back is short-haul, flyable with the narrow-bodies India already has. The long-haul deficit needs an aircraft that has only just been ordered.

> **PIVOT 2**
>
> This case opened on the opposite premise, that Gulf carriers were eating India's market and the wide-body order was a rescue. The trend reversed it. The finding only surfaced after a `GRAND TOTAL` row worth 17.53M passengers was found being counted as a foreign airline. How this analysis changed

Indian carriers have taken 9 points of share back since 2015, and are closing o

Share of India international sector passengers by carrier home region. 2020 and 2021 omitted: repatriation flying



Source: DGCA traffic statistics, via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

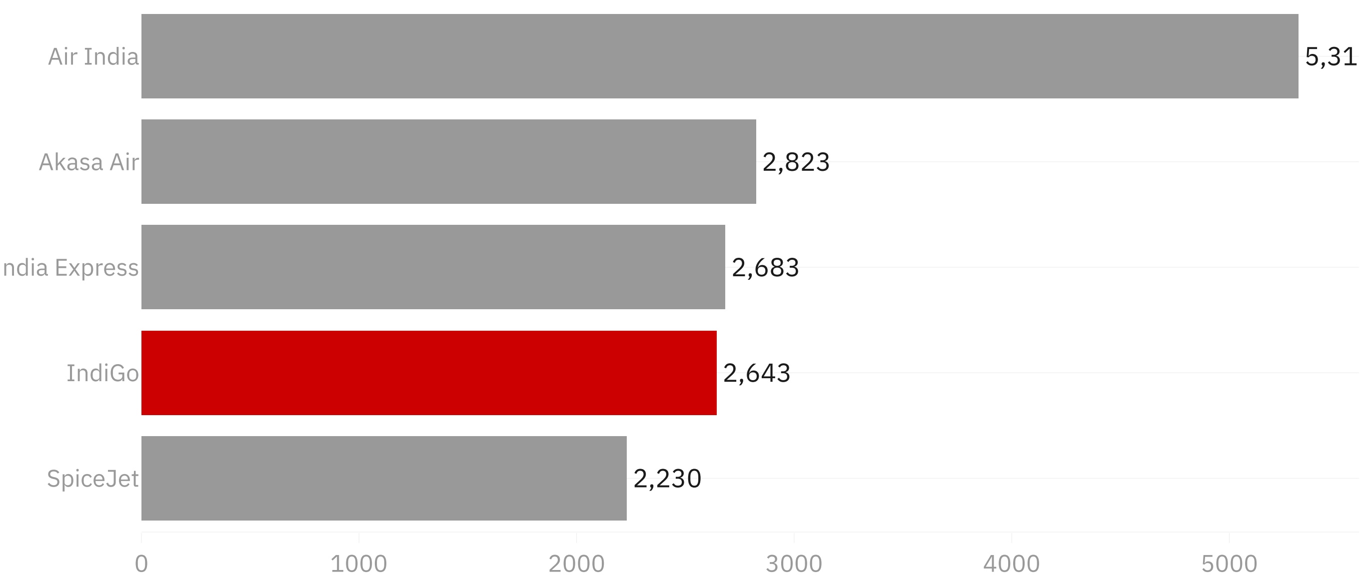
The gap the wide-bodies exist to close

In 2025 IndiGo carried more international passengers than Air India while flying barely half the distance per passenger. Its international network averages under 2,700 km, which is the Gulf and Southeast Asia. Air India's averages over 5,300 km.

Neither airline is doing anything wrong. They are simply in different businesses, and the wide-body order is IndiGo buying its way into the second one.

IndiGo flies more international passengers than Air India, over half the distanc

Average international stage length, km, 2025. Air India runs a long-haul network, IndiGo a Gulf and Southeast Asia



Source: DGCA traffic statistics, via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

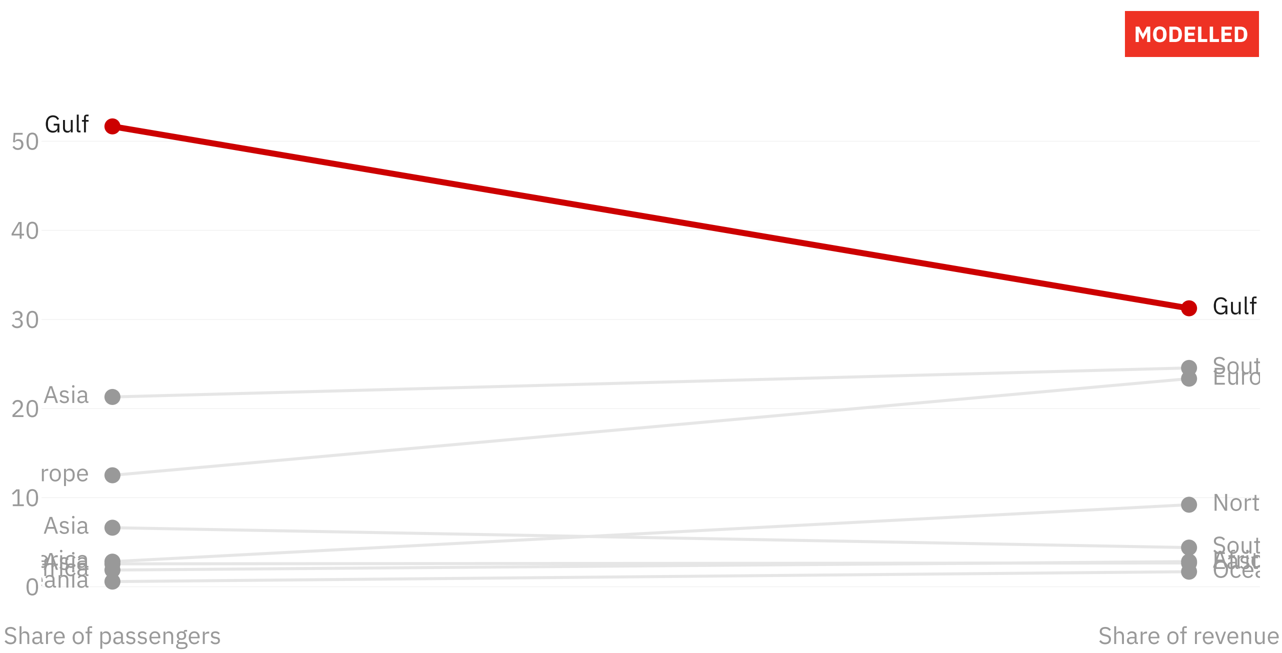
Half the passengers, a third of the money

The Gulf is 52 per cent of India's international passengers. Weight the same traffic by distance flown, and its share of revenue falls to 31 per cent. A Dubai sector is about 2,200 km; a New York sector is about 11,700 km, and revenue scales with the distance.

No margin assumption enters this chart. It is passenger counts, great circle distances and one published yield, so it stands even if you reject the profit model that follows.

Weighting by distance flown moves the Gulf from half the market to a third of i

Percent of India's international corridors, 2024. Passenger counts are DGCA; revenue is passengers x reference gre



Source: Passengers: DGCA international country table, 2024. Distances: great circle from OurAirports (CC0). Yield and margin ancl

Where the profit actually sits

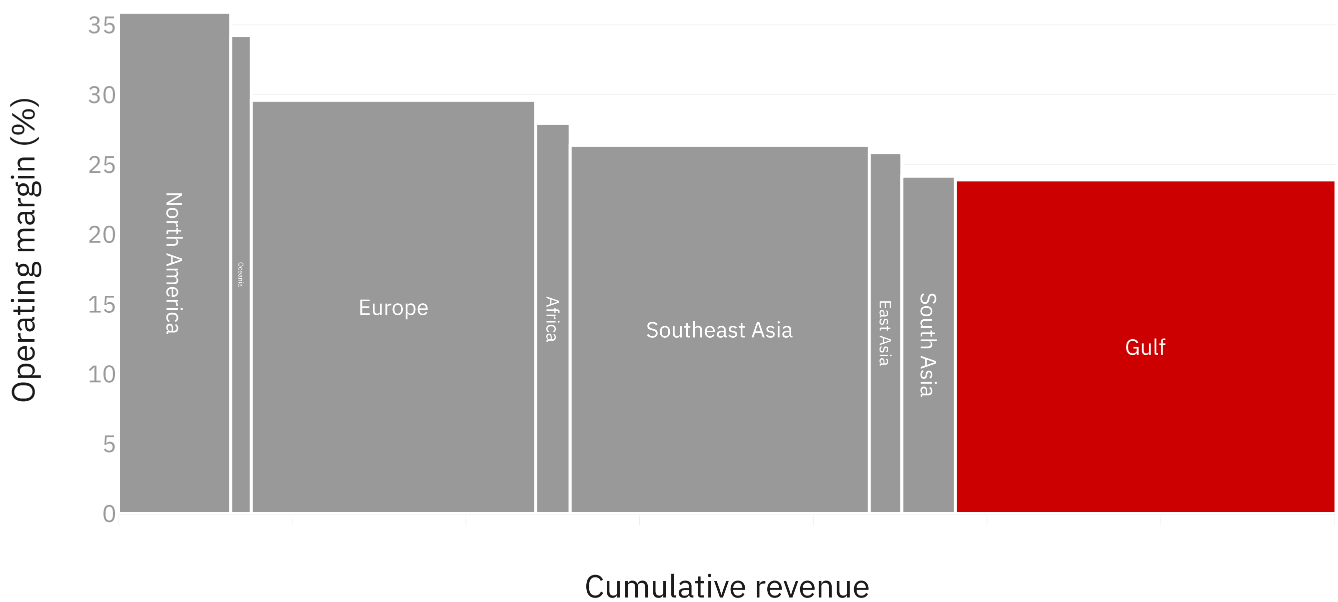
Width is revenue, height is margin, so each block's area is that corridor's profit. The Gulf stays the widest block because volume is real, but the long-haul corridors sit above it, and that height is what a wide-body buys access to.

The margin axis is modelled, and labelled as such, because no Indian carrier publishes a margin split by corridor. It is anchored so the revenue-weighted mean equals IndiGo's 27.3 per cent FY2026 EBITDAR margin excluding forex. IndiGo reported 17.8 per cent including it. Both are stated, because the difference is a treasury outcome on dollar lease liabilities, not a route one.

The Gulf is 52% of passengers but 31% of revenue, a 20 point gap that wide-bo

Width is a revenue proxy (passengers x reference great circle from Delhi x IndiGo's 5.06 INR per RPK); height is a r

MODELLED



Source: Passengers: DGCA international country table, 2024. Distances: great circle from OurAirports (CC0). Yield and margin ancl

And the corridor we are recommending is the tightest one

Unit cost falls as sectors lengthen, because the cost of a departure is spread over more seat kilometres. Scale IndiGo's published unit cost across the corridors and ask a different question: how far could fares fall before each stops covering what it costs to fly?

The Gulf has the least room of any corridor, and on these numbers it does not cover its cost today. Europe could absorb a fifth off the fare and still clear.

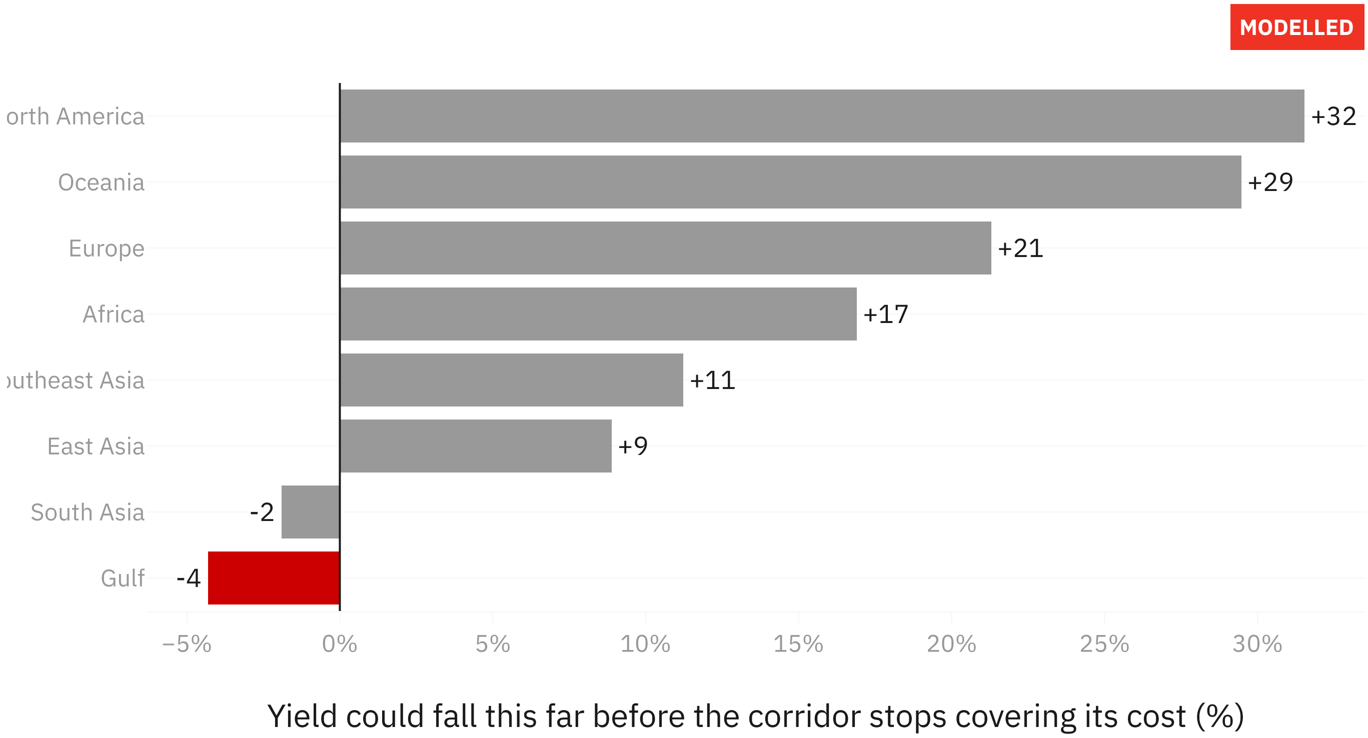
This chart is one of the three findings that moved the recommendation off "Gulf first". It does not say the Gulf stops mattering: the corridor is still where the traffic and the contested revenue are. It says the aircraft should not be pointed at it, because the sectors are too short to carry their own cost and there is no entitlement to fly more of them anyway.

This corroborates the chart before it from an unrelated direction. The profit pool models margin upward from an EBITDAR anchor; this scales cost downward from a published CASK. They share nothing but the corridor distances, and they rank the corridors the same way. A test asserts they keep agreeing.

> **PIVOT 1**
>
> This chart, the bilateral ceiling and the capacity arithmetic together moved the recommendation. It was "reclaim the Gulf corridor first" until three separate lines of evidence said the aircraft cannot go to Gulf airports: no entitlement to fly them on, the worst unit economics on the map, and an order book sized for a network a quarter longer than this one. How this analysis changed

The Gulf has the least room to absorb a yield decline of any corridor, because s

Headroom against IndiGo's achieved 5.06 INR per RPK, at the 81% load factor Indian carriers fly internationally. U



Source: Unit costs and yields from IndiGo FY2026 and Emirates 2025-26 primary filings, verified in data/manual/assumptions.csv

And there is a second deck nobody has counted

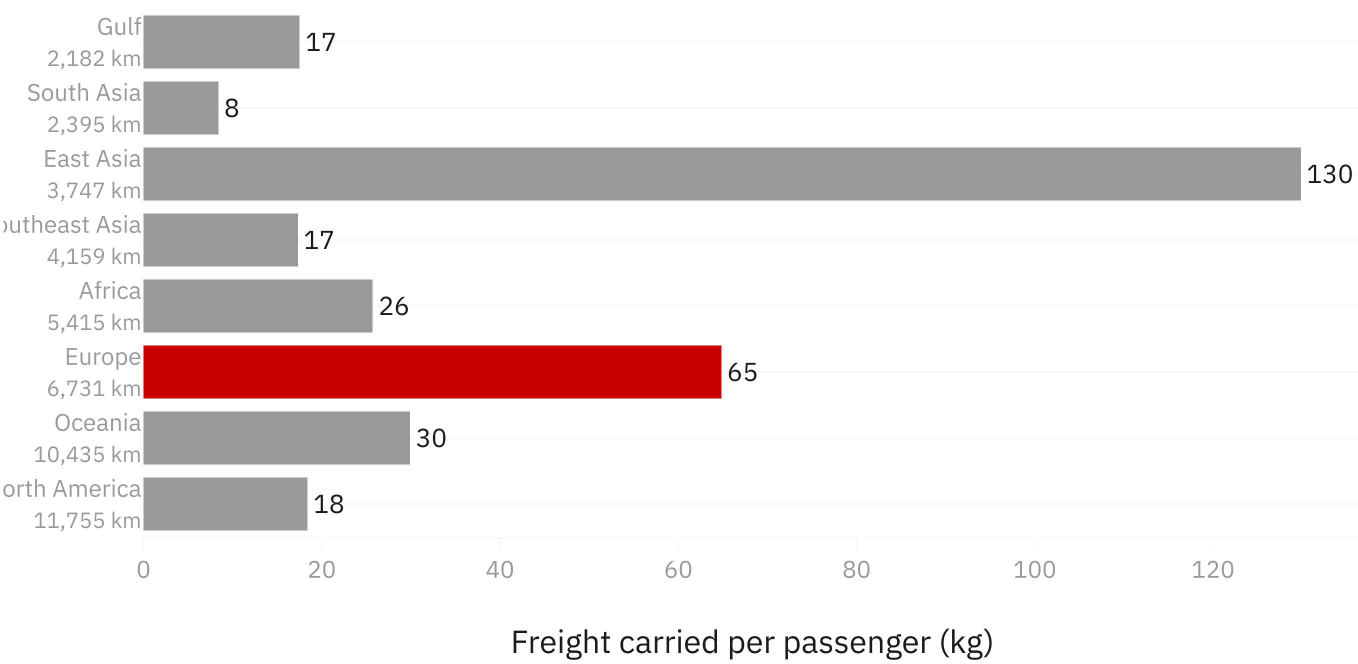
Every wide-body carries freight under the floor, and this case has ignored it until now. Europe moves 65 kilogrammes of it per passenger against the Gulf's 17, so the corridor being recommended carries nearly four times the cargo of the one being stepped back from.

Resist the obvious conclusion. This is not a long-haul effect: the correlation between sector length and freight per passenger is minus 0.10, which is nothing. North America is the longest corridor on the map and sits near the bottom. Belly freight tracks what two economies trade, not how far apart they are, and a chart ordered by distance shows that better than one ordered by freight.

East Asia is the number to look at twice: 130 kilogrammes per passenger on a corridor carrying two million people, the thinnest passenger market here with the densest freight. Nothing else in this case examines it. No revenue is claimed anywhere on this chart, because no Indian carrier publishes a freight yield.

Europe carries 3.7 times the Gulf's freight per passenger, and distance does nc

Corridors ordered by sector length, shortest at the top. The correlation between the two is -0.10, so belly freight tra



Source: Freight and passengers from DGCA international country statistics via github.com/Vonter/india-aviation-traffic (ODbL); co

The market being bought into

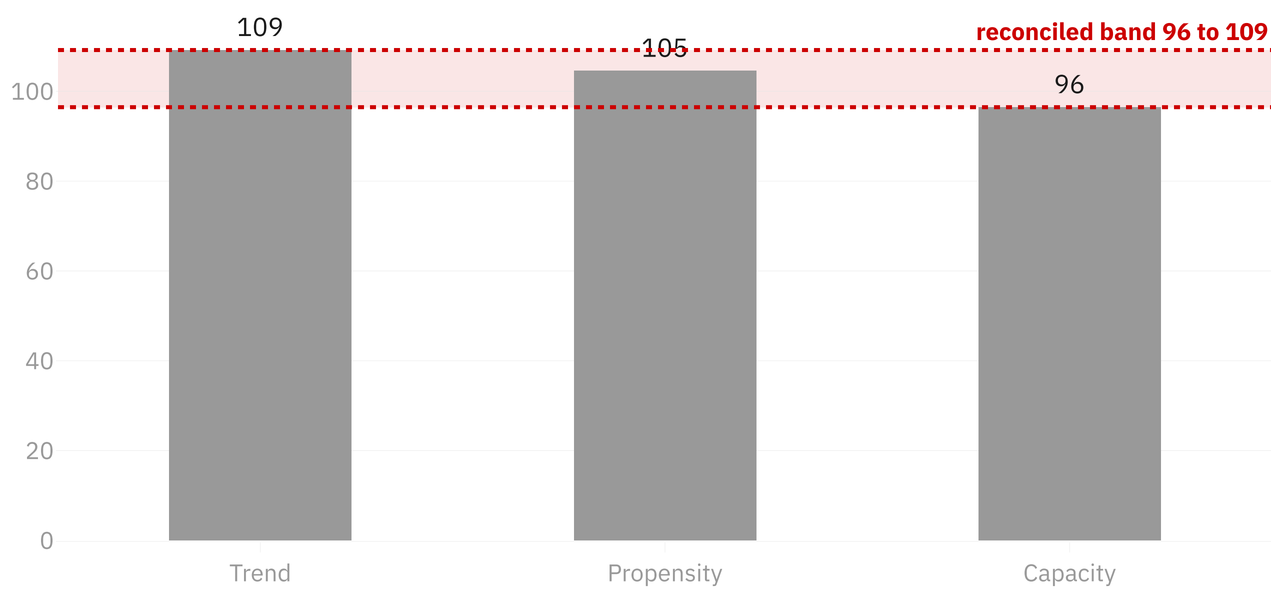
Three methods, extrapolating India's own traffic history, fitting air travel against income across peer countries, and counting the seats the announced order books can actually fly, put the 2030 international market between 96 and 109 million passengers, up from 78 million.

The capacity method is the low leg, and it was withheld for most of this project's life because its inputs were not verified. They now are, from the manufacturers' own airport planning manuals and IndiGo's annual report. The band is reported, never averaged: the spread between the legs is the useful output.

> **PIVOT 5**
>
> Note which direction verifying the gated numbers moved the answer. The new leg is the low one, so the band widened downward and the recommendation got harder to argue, not easier. A gate that only ever unlocks good news is not a gate. How this analysis changed

India's international market reaches 96M to 109M passengers by 2030, up fror

Millions of international sector passengers in 2030, against 78.0M in 2025. The band is reported, never the average



Source: DGCA traffic statistics via github.com/Vonter/india-aviation-traffic (ODbL); World Bank Open Data (CC BY 4.0). Pulled 202

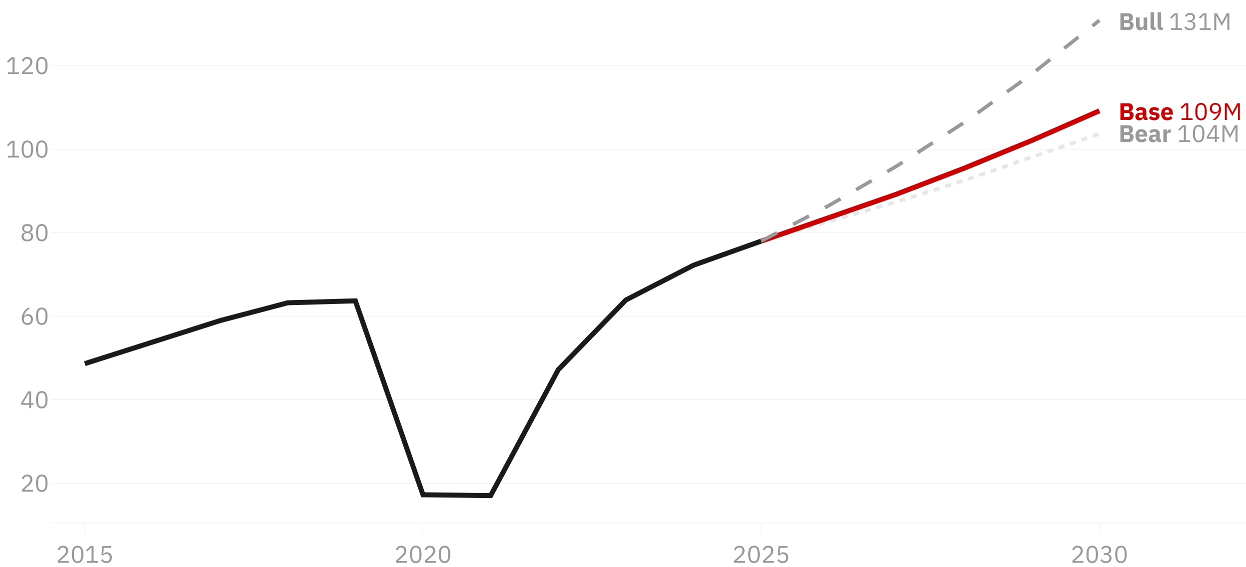
Even the pessimistic case needs the aircraft

Three growth paths, anchored on rates India has actually recorded rather than chosen for symmetry. The bear case is the slowest sustained three-year stretch in the clean data; the bull case is capped at 12 per cent, because beyond that the constraint stops being demand and becomes how fast aircraft, crew and slots can be brought on.

The spread between them is 27 million passengers. Every path requires materially more long-haul capacity than exists today.

Even the bear case adds 26M international passengers by 2030

India international sector passengers, millions. Growth rates anchored on observed history, not chosen for symme



Source: DGCA traffic statistics via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

The aircraft are not the constraint. The network is

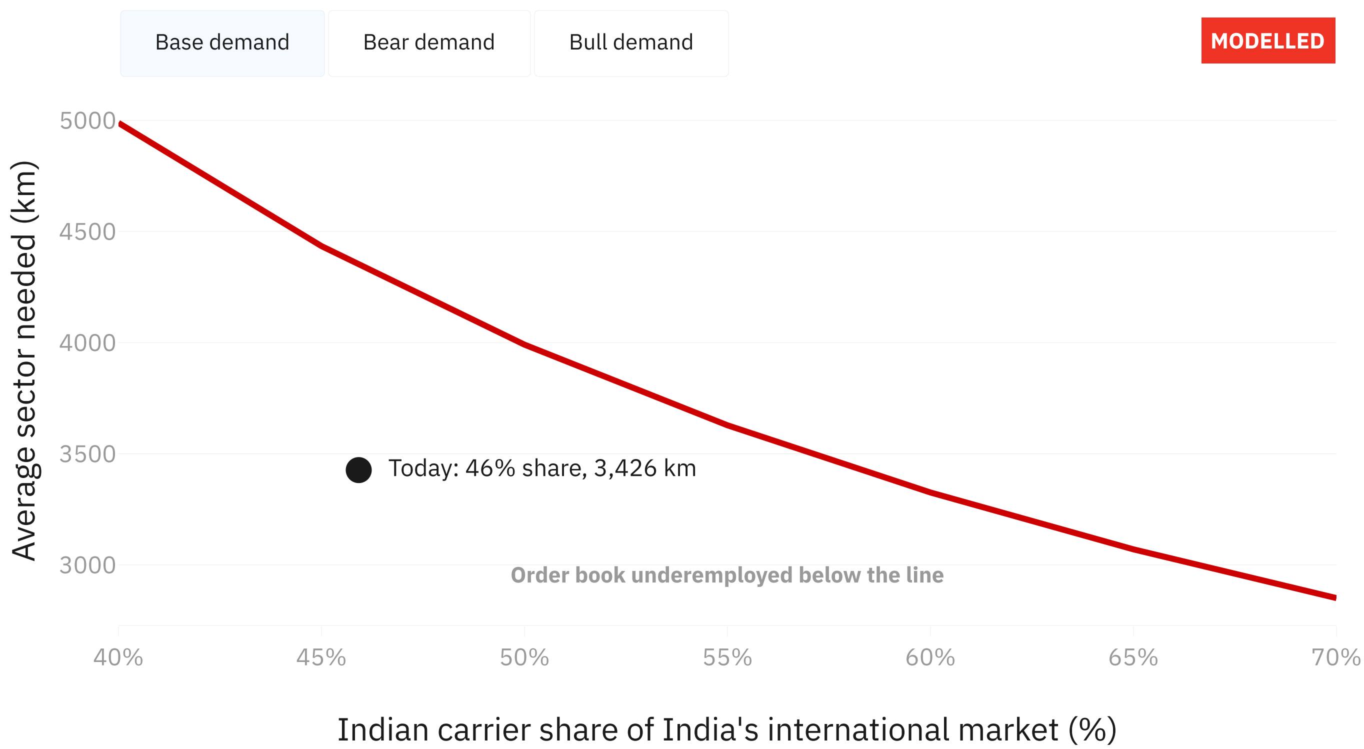
Count what the firm order books can actually fly and the answer inverts. The 140 wide-bodies on order carry roughly twice the capacity Indian carriers need to hold their current share of a 2030 market. On today's network shape they would fly half empty of purpose.

They only clear if the average sector lengthens by about a quarter, from 3,400 kilometres to nearer 4,300, or if Indian carriers take share well past parity, or some mix of the two. That is the recommendation restated in capacity terms: these aircraft are not bought to carry more of the same traffic, they are bought to carry it further.

Block speed and sector length here are computed from DGCA's own aircraft-kilometre and aircraft-hour columns, not assumed. Air India blocks at 698 km/h across a 5,316 km network and IndiGo at 656 km/h across 2,643 km, which is the physical relationship you would expect and a reason to trust the columns.

The order book clears only if the average sector lengthens about 27%, or share

Combinations that exactly absorb 119bn ASK of firm wide-body order on top of today's 154bn, against a 111M pass



Source: Capacity, stage length and block speed computed from DGCA carrier operating statistics via github.com/Vonter/india-avia

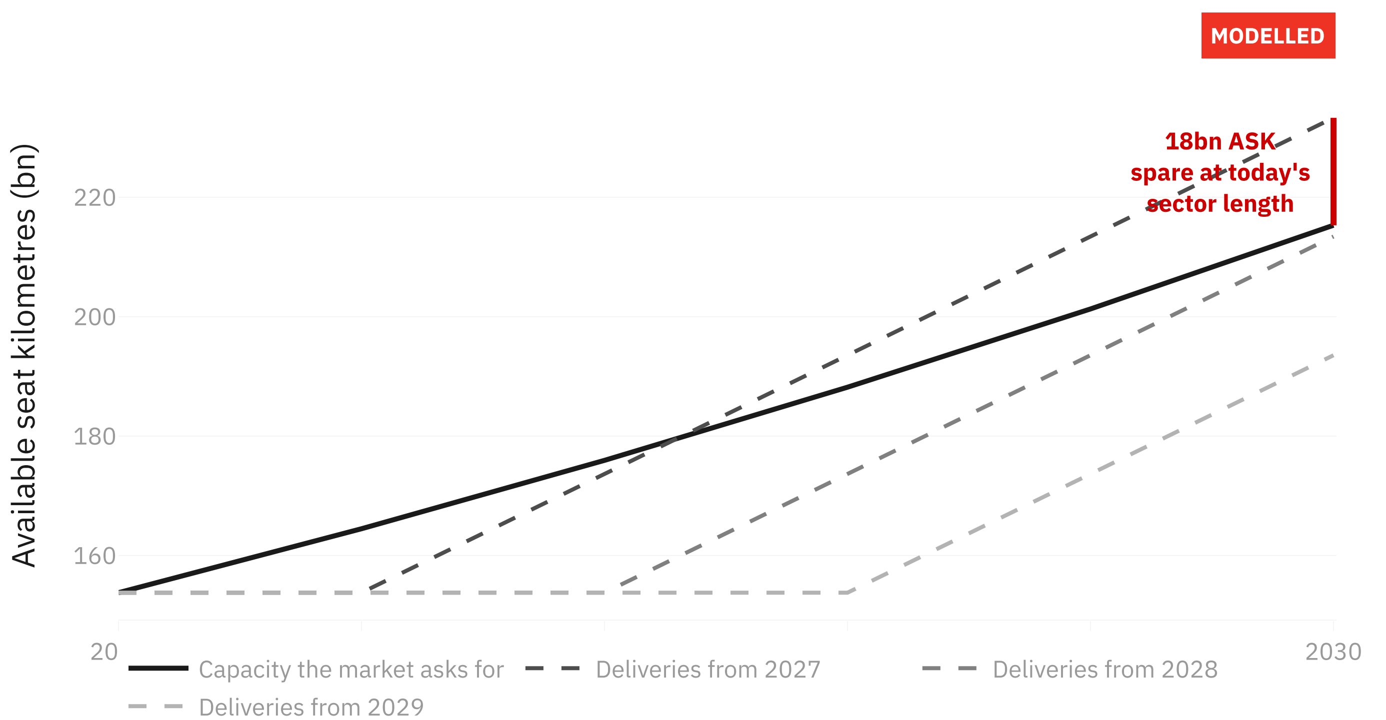
Timing changes when it lands, not whether it is enough

No primary source says when the deliveries begin. The Airbus release that confirms IndiGo's sixty firm A350s states no schedule at all, so no start year is asserted anywhere here. Three plausible starts are run instead and the spread is the output.

The shape is the same in all three: a gap opens in the bridge years, widest between 2026 and 2028, and closes as aircraft arrive. Slipping the start moves the whole delivery curve to the right without changing where it ends up, which is why the phasing question is about what to fly in the meantime rather than about how much was ordered.

Delivery timing changes when the capacity lands, not whether it is more than l

Indian carrier international ASK, base demand case, share held at 46% and sector length held at today's 3,426 km.



Source: Capacity, stage length and block speed computed from DGCA carrier operating statistics via github.com/Vonter/india-avia

The cost problem is a currency problem

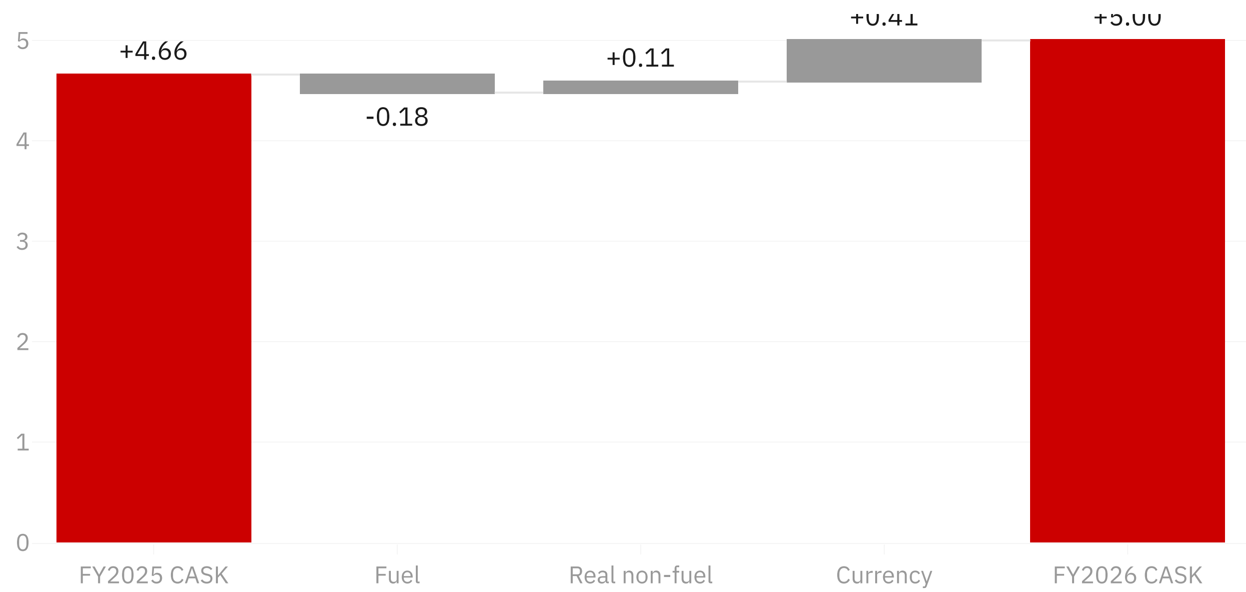
IndiGo's unit cost rose 4.66 to 5.00 rupees per available seat kilometre in FY2026, and the headline reads as an operating collapse. Bridge it and the picture inverts: fuel fell 0.18, genuine non-fuel inflation was 0.11, and currency added 0.41. The currency effect alone is larger than the entire net increase.

This is the same correction as the retracted margin claim in the methodology, made at the unit cost line instead of the margin line. A dollar-denominated lease book is not an operating problem, but it is a real one.

> **PIVOT 4**
>
> This exhibit exists because an earlier version of this page claimed IndiGo's operating margin had halved, 22.3 to 14.0 per cent. It had not. The number came from a convention the company does not publish, the real margin improved, and the claim was withdrawn. The pressure is real but it sits in unit cost, which is what this chart measures. How this analysis changed

Currency added +0.41 to IndiGo's unit cost, more than the entire +0.34 increas

Rupees per available seat kilometre, FY2025 to FY2026. Real non-fuel cost inflation was +0.11. Reading the headlir



Source: IndiGo Q4 and FY2026 results press release; IOCL aviation fuel prices; FBIL USD/INR reference rate. All rows verified in d

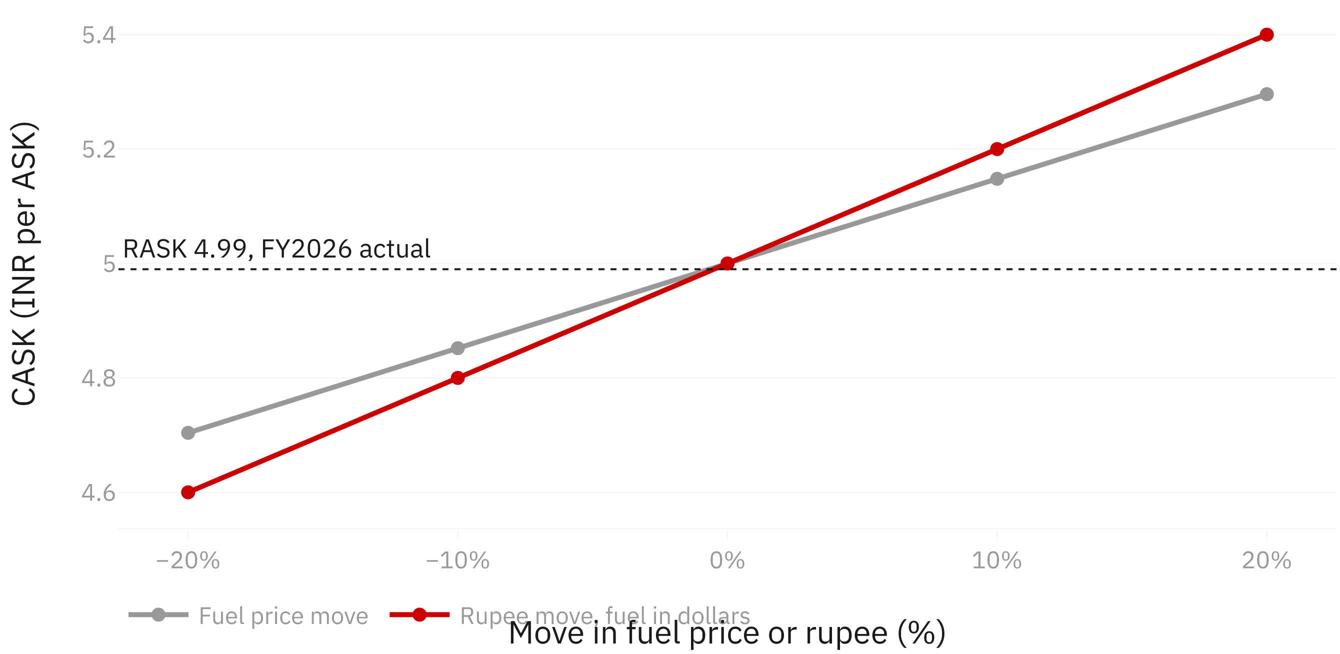
Which shock hurts more

Between 10 and 40 per cent of IndiGo's unit cost moves with the rupee, depending on how much fuel is bought at the dollar international price rather than the rupee domestic one. India publishes no split, so the exposure is reported as a band.

At the top of that band a rupee move costs more than the same move in fuel. And every line starts above breakeven, because FY2026 opened with unit cost at 5.00 against unit revenue of 4.99. Long-haul aircraft are bought into that, not out of it.

A rupee move costs more than the same fuel move, and every path starts above

CASK against FY2026 actual RASK, held flat because pass-through has not been measured here. FY2026 already o



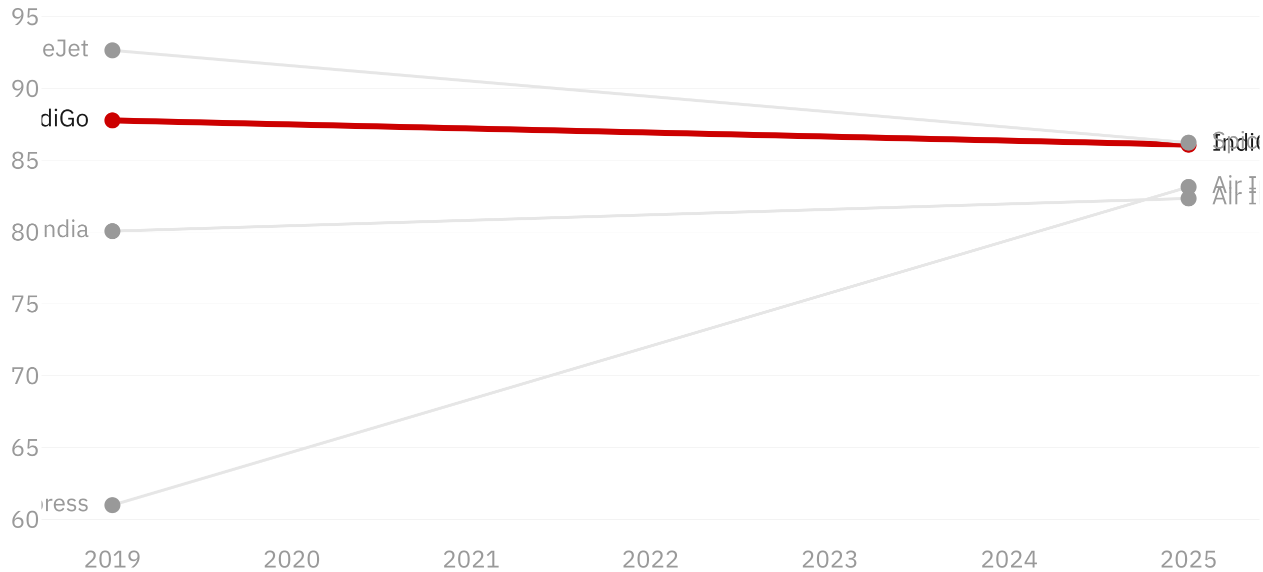
Source: IndiGo Q4 and FY2026 results press release; IOCL aviation fuel prices; FBIL USD/INR reference rate. All rows verified in d

Demand is not the constraint

Domestic load factors have recovered past their pre-pandemic level. The aircraft that exist are full. The question is what the next hundred fly, and where.

Domestic load factors have recovered past their pre-pandemic level

Scheduled domestic passenger load factor, per cent



Source: DGCA traffic statistics, via github.com/Vonter/india-aviation-traffic (ODbL). Pulled 2026-08-15

Two statistics agencies, opposite ends of the same routes

Every figure above rests on Indian government data. Before relying on it, the same routes were measured from the European end using Eurostat, which has no knowledge of the Indian series. Across the seven countries both agencies cover cleanly they agree to 2.6 per cent. This exercise is run on **2024**, the last year both agencies publish complete, while the figures above are 2025. The reconciliation validates the source, not one year's numbers.

COUNTRY	DGCA, INDIA END	EUROSTAT, EUROPE END	GAP
Finland	149,508	149,551	0.0%
France	964,079	955,339	-0.9%
Switzerland	340,372	343,583	0.9%
Germany	1,744,685	1,766,890	1.3%
Netherlands	664,238	673,438	1.4%
Poland	108,223	109,731	1.4%
Denmark	95,986	99,036	3.2%
Italy	222,689	305,303	**37.1%**
Total	**4,289,780**	**4,402,871**	**2.6%**

The cross-check found something

Italy is the one outlier, and the entire gap is a single route. Milan to Delhi agrees between the two agencies to 1.6 per cent. Eurostat reports 171,942 passengers on Rome Fiumicino to Delhi in 2024; DGCA lists no Rome to Delhi pair at all. It is not a naming mismatch, since DGCA uses "ROME" elsewhere in the same file for Rome to Amritsar.

No free source settles which agency is right. The route is therefore quarantined, excluded from every figure that would depend on one of them being correct, and reported here with both numbers rather than resolved to whichever was more convenient.

What to do, and what would have to be true

Europe first, North America second, Gulf capacity roughly flat. The corridor stays the centre of the case; the aircraft point elsewhere.

The options

Time to capacity and capital intensity are judgements, so they are words rather than an invented score. The columns that are computed say so.

OPTION	TIME TO CAPACITY	CAPITAL	WHAT WOULD HAVE TO BE TRUE
Own wide-bodies on long-haul	2027 at the earliest	High	Deliveries arrive on a schedule no primary source states, and Europe direct yields hold within about a fifth of today's realisation
Own wide-bodies on Gulf routes	2027 at the earliest	High	A bilateral renegotiation India has refused for a decade, *and* unit economics that are currently negative turning positive
Lease or damp-lease bridge	Immediate	Low, recurring	Lease rates leave headroom over the corridor's cost. Not quantifiable here: rates are paywalled trade press and no assumption row can clear
A321XLR on thin routes	Near term	Medium	The corridor sits inside 8,700 km, which covers Europe and excludes North America *(computed)*
Codeshare or joint venture	Immediate	None	A partner hub is worth more than the margin ceded, which reverses the recapture logic the rest of this case rests on
Do nothing	n/a	None	The connect gap stays with the Gulf carriers, and aircraft already ordered fly at today's sector length

The first row is the recommendation because it is the only option that is simultaneously available, unlike a Gulf bilateral; affordable in unit cost terms, unlike Gulf sectors; reachable, unlike North America by narrow-body; and capable of absorbing capacity already contracted for.

Where it fails

Likelihood and impact are judgements. What matters more, and is why this is a table and not a scatter plot, is the last two columns.

RISK	IMPACT	WHAT WOULD FALSIFY IT	WHAT TO WATCH
Delivery slip	High	A published Airbus schedule for the IndiGo book, which does not exist	Airbus quarterly deliveries
Rupee depreciation	High	Sustained USD/INR stability	FBIL daily reference rate. Currency added +0.41 to FY2026 unit cost, more than the entire net rise

RISK	IMPACT	WHAT WOULD FALSIFY IT	WHAT TO WATCH
Gulf carriers cut fares on competing routes	High	Gulf capacity discipline as their own fleets tighten	Fares on India-Europe against India-Gulf-Europe itineraries
Europe yields fall past the headroom	High	Yields holding within a fifth of today's realisation	IndiGo quarterly yield, which is published
Bilateral expansion re-floods the Gulf	Medium	India holding its refusal, as it has since 2014	Ministry of Civil Aviation announcements
The connect prize is smaller than modelled	High	A published origin-destination table, which IATA sells rather than publishes	Any free O-D release, or a second agency covering India-Gulf

The most likely reason this is wrong

The last row, and it is a data risk rather than a market one. Only **5.6 per cent** of India's international traffic is cross-checked against a second agency. The Gulf, which carries half of it, has no equivalent open source at all, so the cross-checks that caught five of the six errors below cannot be run on the corridor this case turns on.

How this analysis changed

Six times, evidence turned the work against what it had been assuming. Once it changed the recommendation itself. Once it reversed the premise of the whole case. Once it forced a claim off this page. Once it found a wrong answer that every test in the repo had passed. They are here because a case that arrives in a straight line was either trivial or is hiding something.

1

"Gulf first" became "compete with the Gulf hubs, do not fly to them"

Believed, and it was this page's headline: reclaim the India-Gulf corridor first, defend long-haul second. It is half of India's international traffic and four times the entire direct Europe market, so that is where the wide-bodies should go.

Evidence: the traffic is there, the aircraft cannot be. India-Dubai runs at 88.8 per cent of its treaty limit and Abu Dhabi at 70, and the room the two leave together absorbs about four per cent of the order book. The Gulf has the worst unit economics of any corridor. And the order book would add 78 per cent to Indian carriers' international capacity when holding share needs about half that, clearing only if the average sector lengthens about a quarter.

Changed: the recommendation, and this page's headline with it. The corridor stays the centre of the case, because that is where the traffic and the contested revenue are. The aircraft point elsewhere. Of 39.7M Gulf passengers roughly 31.2M are genuinely point-to-point and already well served; about 8.5M are connecting onward, and that traffic is won by flying past the Gulf rather than to it.

What it should have caught earlier: branch 4.3 of the hypothesis tree already said incremental Gulf capacity "has to go long-haul, or to Gulf points with slack, or wait on a renegotiation India has so far refused". That sat in the tree and never reached the headline. The analysis was ahead of the write-up.

2

"India is losing its own market" became "India is winning it, just not in long-haul"

Believed: Gulf carriers are eating India's international market and the wide-body order is a rescue. It is the version of this story that gets written most often.

Evidence: the opposite. Indian carriers went from 37.0 to 45.9 per cent of India's international sector passengers between 2015 and 2025, while Gulf carriers fell from 32.7 to 26.2. The trend only surfaced after a `GRAND TOTAL` row worth 17.53M passengers was found being counted as a foreign airline.

Changed: the storyline was rewritten around the finding rather than against it, and the argument got sharper. The share taken back is short-haul, flyable with narrow-bodies already owned. What remains is a range constraint, which is a better case for a wide-body order than "we are losing".

3

A wrong answer that passed all 72 tests

Believed: the Gulf-hub flow diagram was right, because the suite was green.

Evidence: the code looked for `ABU DHABI` and `RAS AL KHAIMAH`. DGCA writes `ABUDHABI` and `RAS AL-KHAIMAH`. Exact matching missed both, filing 5.0M passengers a year, about a fifth of the entire Gulf hub flow, under "everywhere else, direct". Shares still summed to 100, flows were still well formed, nothing was null. Every property the suite checked was true of an answer that was wrong.

Changed: matching now goes through a normalised key, which fixes the whole class rather than the two names that happened to be noticed. Three tests were added that check the thing which actually failed. The bug was found while chasing a bilateral seat number, not by any test.

4

A margin claim published here, then withdrawn

Believed, and stated on this page: IndiGo's operating margin had halved from 22.3 to 14.0 per cent, and wide-bodies would have to be funded out of that squeeze.

Evidence: the figures came from an aggregator's "operating profit" line. ₹18,050 cr is not something IndiGo publishes, so it could never have been verified against a primary source. Against the company's own filings the margin improved, 26.3 to 27.3 per cent ex forex.

Changed: the claim was withdrawn and the retraction kept rather than deleted. The real pressure was found where it actually sits, in unit cost, which became the cost bridge exhibit. The symmetric error was then guarded against: IndiGo reported 17.8 per cent, 27.3 is ex forex, and both now appear wherever either does. The whole assumption gate exists because of this.

5

Verifying the gated numbers made the recommendation harder to argue

Believed: the capacity leg of the market sizing was blocked on paperwork. Find the seat counts, unblock it, move on.

Evidence: sourced from the Airbus and Boeing airport planning manuals and IndiGo's own block hours, the new leg came in as the *low* one, widening the band downward. Current values are on the sizing chart above; they are deliberately not repeated here, because this entry is about what the gate did.

Changed: nothing was relaxed to make it work, and note the direction. The gate opened onto a result that made the case harder. A gate that only ever unlocks good news is not a gate.

6

A widely quoted utilisation figure that did not survive arithmetic

Believed: IndiGo runs its fleet at roughly 13 block hours per aircraft per day. The figure appears in trade coverage and had been carried in this project's own notes.

Evidence: it cannot be reconciled with published block hours. 1,619,570 hours in FY2026 at 13 hours a day requires 100 of 441 aircraft grounded. Reported groundings were in the 40s, which gives 11.07. The plausible range is about 10.5 to 11.7.

Changed: the figure was retired. The capacity leg runs on the owned-fleet basis of 10.06 hours and the chart names that basis. The replacement was then cross-checked in its own right: DGCA independently reports 1,614,608 hours for the same carrier and year, 0.31 per cent apart.

What these have in common

Four of the five were found by cross-checking one source against another, or one number against its own arithmetic. None were found by inspection, and none by the test suite. That is the argument for the reconciliation above being an exhibit rather than an appendix: disagreement is the only mechanism here that reliably surfaces a wrong answer.

The corollary is uncomfortable and worth stating. Only 5.6 per cent of India's international traffic is cross-checked against a second agency. The Gulf, which carries half of it, has no equivalent open source, so the checks that caught these five cannot be run there.

Method and sources

Where the numbers come from

- **DGCA traffic statistics**, via the ODbL mirror at `github.com/Vonter/india-aviation-traffic`. Five datasets, fresh to May 2026.
- **Eurostat** air transport by airport pair, for the European end of India-Europe routes.
- **World Bank Open Data** for income, population and air travel propensity across twelve peer countries.
- **OurAirports** (CC0) for airport reference data.

Every figure on this page is recomputed from those sources by `scripts/refresh.py`. None are typed in by hand.

What this data cannot tell you

- **Sector, not origin-destination.** DGCA counts the India to foreign point leg. A Delhi to Dubai to London passenger appears under the UAE. This is why the Gulf reads 51 per cent here against roughly 40 per cent in IATA's true O-D figures, and that gap is the connecting traffic.
- **Yields are not published.** DGCA publishes no fares and Air India is unlisted, so it files nothing. Every revenue figure is therefore gated: the code refuses to use an assumption until it has been checked against a primary source.
- **Two sources were dropped** after one attempt each rather than scraped unreliably: BTS T-100 and Indian Oil's fuel prices. The United States arm is measured from the India side only, and it says so.

The written case

The long-form documents behind this page. They render on GitHub, alongside the code that produces every figure here.

Storyline
The brief, the recommendation and the SCQA that supports it

Recommendation
The option menu, the phased roadmap, what would have to be true, the risk register and the leading indicators

Hypothesis tree
The case decomposed before any analysis ran, with each branch marked answered, partial or blocked

Pivot log
The five entries above, in full, each citing the commit where it happened

Survey design

A conjoint instrument for the one number this case cannot verify. Designed, not fielded, and the coverage report still counts it as a gap because of that

Methodology

What was done, what it rests on, what it cannot tell you, and the retraction

Data dictionary

Every field, its source, pull date and reliability grade

Coverage

This case mapped against the job posting, with the gaps stated rather than stretched

One trap worth naming

DGCA publishes distance columns in thousands while passenger counts are raw, and the files say so nowhere. Taken at face value the average domestic passenger flies 0.98 km. The correction was confirmed three ways before it was applied: the scale reproduces India's published 163.8bn domestic RPK; computed load factor matches DGCA's own published column to 0.25 percentage points; and the aircraft-kilometre and passenger-kilometre columns independently give 588 and 589 km per departure for the same month.

Built as a portfolio case study in the style of Bain Capability Network Advanced Manufacturing & Services commercial aviation work. Analysis in Python, charts in Plotly, no PowerPoint and no Excel anywhere in the pipeline. Mekko builder adapted from Vizro under Apache-2.0; see `NOTICE`.